OCI Optical Link Budget — GEN1 (53.125 GBd) and GEN2 (106.25 GBd CPO) Unified Engineering Report

Scope. Bottom-up, OMA-domain link budgets per DWDM channel for the 200G OCI line interface at an internal pre-FEC BER target of 10^{-12} ($Q = 7.035$), covering the GEN1 MSA-rate link (53.125 GBd NRZ) and the GEN2 co-packaged-optics (CPO) link (106.25 GBd NRZ, analog SerDes, CTLE-only receiver). The document unifies four analysis canvases and details the methodology and calculations behind every result. All numbers trace to the runnable scripts in this folder (`01`–`06`, see Appendix A) or to a labeled assumption in §7.

0. Glossary

Acronyms / abbreviations

Term	Meaning
OCI	Optical Compute Interconnect
MSA	Multi-Source Agreement
GBd	Gigabaud
NRZ	Non-Return-to-Zero
CPO	Co-Packaged Optics
BER	Bit Error Rate
FEC / KP4	Forward Error Correction / the RS(544,514) code used at 100–200G
OMA	Optical Modulation Amplitude
TIA	Transimpedance Amplifier
SRS	Stressed Receiver Sensitivity
TDEC(Q)	Transmitter and Dispersion Eye Closure (Quaternary, for PAM4)
SEC(Q)	Stressed Eye Closure (Quaternary)
ER	Extinction Ratio
MPI	Multipath Interference
RIN	Relative Intensity Noise
ORL	Optical Return Loss
ISI	Intersymbol Interference
EQ	Equalization
FIR	Finite Impulse Response (filter)
CTLE	Continuous-Time Linear Equalizer
DFE	Decision Feedback Equalizer
CD	Chromatic Dispersion
DJ / RJ / TJ	Deterministic / Random / Total Jitter
DCD	Duty-Cycle Distortion
UI	Unit Interval
PD	Photodiode
MRM	Micro-Ring Modulator
MRR	Micro-Ring Resonator
EIC / PIC	Electronic / Photonic Integrated Circuit
ELS	External Laser Source
SerDes	Serializer/Deserializer
CID	Consecutive Identical Digits
AGC	Automatic Gain Control
LF	Low Frequency
SE	Single-Ended
BT4	Bessel-Thomson 4th-order (reference filter)

Term	Meaning
TP2 / TP3	Test Point 2 / 3 (Tx output / Rx input)
IL	Insertion Loss
SMF	Single-Mode Fiber
DWDM	Dense Wavelength-Division Multiplexing
TT	Typical-Typical (process corner)
SOA	State of the Art

Math symbols

Symbol	Meaning	First used
Q	Personick Q-factor	§2.2
i_n	Input-referred rms noise current (TIA)	§2.3
R	Photodiode responsivity (A/W)	§2.3
Z_T	Transimpedance gain	§2.3, §5
B_n	Noise integration bandwidth, $1.5 \times f_{3\mathrm{dB}}$ convention	§2.3
$H(f)$	Transfer function (generic)	§2.3–2.5
$\mathrm{OMA}_{\mathrm{floor}}$ / $\mathrm{OMA}_{\mathrm{sens}}$	Analytic noise-limited sensitivity	§2.1, §2.3
P_i	Penalty stack line i (dB)	§2.1, §2.6
h_{-1}, h_{+1}, h_{+2}	Pulse-response cursors (pre-, post-1, post-2)	§2.4
τ	Pole time constant	§2.4
$t_{20\text{–}80}$	20–80% transition time	§2.4
ω_z, ω_p	CTLE zero / pole (angular freq.)	§2.5
η	CTLE rms noise-enhancement factor	§2.5
D_{MPI}	MPI discount factor	§2.6
D_{disp}	Fiber dispersion coefficient (ps/nm), CD penalty formula	§2.6
S	MPI reflection-pair sum	§2.6, §3.3
R_t, R_r, R_c	Transmitter / receiver / connector reflectance	§2.6, §3.3
E	Extinction ratio (linear, in MPI formula)	§2.6
σ_{RJ}	RJ standard deviation	§2.6
I_{DK}	Dark current	§2.6
q	Electron charge	§2.6
ε	Crosstalk factor	§2.6
δ	Threshold-offset fraction of swing	§2.6
λ	Wavelength	§2.4 (CD)
N	CID run length (bits)	§5
f_{LF}	Low-frequency cutoff	§5
$f_{3\mathrm{dB}}$	3 dB bandwidth	throughout

Note: D is used for two distinct quantities in the source derivations (MPI discount factor and fiber dispersion coefficient). They are disambiguated above as D_{MPI} and D_{disp} for this glossary, though the body text below still uses the bare D in each local context — read it per §2.6's two formulas.

---

1. Executive summary

The link budget's purpose is to **derive component requirements** — above all, the GEN2 receiver TIA class — not to grade existing parts. Measured TIA characterization data (152 gain/peaking settings of the Ocelot ADFET family, TT corner) is used throughout as **model-calibration input**: it anchors the input-referral convention, exposes the family's noise-versus-bandwidth scaling law, and supplied the group-delay lesson that became a headline spec line.

Targets.

Item	Value
Internal pre-FEC BER target	10^{-12} (Personick $Q = 7.035$)

Item	Value
OCI MSA compliance threshold	2.4×10^{-4} ($Q = 3.49$)
KP4 FEC role at the internal target	Pure margin — designing to 10^{-12} pre-FEC means the FEC carries no traffic-dependent burden

GEN1 (53.125 GBd NRZ, per OCI MSA v1.0).

Item	Value	Calculation
Analytic receiver floor	**−12.93 dBm OMA** (calibrated on the lowest-noise bandwidth-adequate TIA setting: $i_n = 3.17\ \mu$A rms, 29.3 GHz)	C.1
Penalty stack	**2.93 dB**	C.2
Required OMA at the receiver	**−10.00 dBm**	C.3
Closure vs. spec-minimum transmitter	**+0.60 dB**	C.4
Closure vs. realistic transmitter (−3.2 dBm OMA, TDEC ≈ 2 dB)	**+2.30 dB**	C.4
Spec finding	At the spec's −19 dB end reflectances the derived MPI penalty would be 0.51 dB — 2.5× the MSA's 0.2 dB allocation. Meeting ~0.2 dB requires **≤ −24 dB** ends. Because GEN1 and GEN2 share the product line, the ≤ −24 dB requirement is enforced for both generations, ensuring the ~0.2 dB MPI line by construction (GEN1 books 0.24 dB, §3.3)	C.5

GEN2 (106.25 GBd NRZ, CPO).

Item	Value	Calculation
Rate scaling	UI halves to 9.412 ps, and Nyquist doubles to 53.125 GHz	C.6
Derived TIA class (§5)	$f_{3\mathrm{dB}}$ 50–64 GHz, $i_n \le 4.0\ \mu$A rms (≤ 14 pA/√Hz average density over $B_n = 1.5 \times f_{3\mathrm{dB}}$), peaking ≤ 1 dB, group-delay ripple ≤ 3 ps	C.7
Target-class TIA floor	58 GHz, 4.5 μA → floor **−11.41 dBm**	C.8
Penalty stack (typical Tx corner)	**4.07 dB**	C.9
Closure margin, typical Tx corner	**+1.34 dB** at Tx OMA −3.5 dBm	C.10
Closure margin, fast / slow Tx corners	+1.67 dB / +0.79 dB	C.10

GEN2 budget waterfall (typical corner). Delivered power: Tx OMA −3.5 dBm − 2.5 dB link IL = **−6.0 dBm at the Rx**. Required power, built up from the receiver floor: −11.41 dBm (floor) + 1.16 (shot/RIN) + 0.21 (MPI) + 1.15 (ISI+EQ) + 0.04 (CD) + 0.95 (jitter) + 0.36 (crosstalk) + 0.21 (threshold) = **−7.34 dBm required** (lines shown rounded; the unrounded stack is 4.07 dB — C.9). Margin = −6.0 − (−7.34) = **+1.34 dB**. (Note the direction: penalties stack *on the floor* to form the required OMA; they are not subtracted from the delivered −6.0 dBm.)

Bottom line: feasible, but specification-sensitive. The GEN2 link closes at every modeled corner, but +1.34 dB typical / +0.79 dB slow-Tx is workable rather than comfortable — the worst-everything device cell reaches +0.16 dB (§6.1), and several independent Low-confidence assumptions (table below) can each move the margin by 0.2–1 dB. The doubled rate does not break the optical budget; it moves the problem into the receiver implementation (the §5 TIA class) and into validating the assumptions below.

Device trade study. Against user-provided device brackets, the **preferred baseline architecture is a no-FIR 60 GHz driver with the 60 GHz / 4–5 μA TIA (option B)**: **+1.63 dB** typical margin, degrading to +0.16 dB only in the worst-everything corner (5 μA noise and a 40 GHz MRM) — degraded but never broken, though that corner now leans on the Tx OMA relief valve. A 3-tap slice-DAC Tx FIR buys no ISI improvement on this short, clean channel (the joint optimizer drives the taps to zero) and costs 0.27 dB of slice-DCD jitter. **A single post-cursor tap is retained as a design option pending measured MRM EO response** — MRM detuning peaking is the one impairment a Tx FIR fixes that the CTLE cannot (§6.3, §8).

Key derived requirements.

- GEN2 TIA table (§5, headline above)
- Noise integration bandwidth convention: $B_n = 1.5 \times f_{3\mathrm{dB}}$, both generations (§2.3)
- End reflectance ≤ −24 dB at both Tx and Rx (both generations — shared product line)
- Tx 20–80 % transition time ≤ 5.6 ps (0.60 UI) hard / 4.2 ps (0.45 UI) target
- Microbump ≤ 25 fF
- Jitter budget: RJ ≤ 141 fs rms, DJ ≤ 1.32 ps (TJ@1e-12 = 3.30 ps)
- TDEC ≤ 1.8 dB

Status and confidence of the load-bearing inputs. The requirements above are only as strong as what they stand on. Reading guide: *Measured* = taken from characterization data in the repos; *Derived* = computed by the scripts from the §2 framework; *Assumed* = judgment call, no supporting measurement; *Required* = design decision this report enforces; *Model result* = output of the budget, inherits the confidence of its inputs. Full provenance per §7 and `Methodology_Provenance.md`.

Item	Status	Confidence
GEN1 design-point TIA (i_n 3.17 μA, 29.3 GHz), R = 0.876 A/W	Measured (152-setting tables)	High
Family noise-vs-bandwidth scaling law (f^2-dominated)	Measured + regression fit	High
GD-ripple failure mode (12.5 ps → h_{-1} ≈ 0.48)	Measured	High
TIA noise ceiling ≤ 4.0 μA	Derived (stack inversion, §4.4)	Medium — hinges on the B_n convention and the stack's assumed lines
TIA bandwidth window 50–64 GHz	Derived (sensitivity sweep, §5)	Medium
GD ripple ≤ 3 ps / peaking ≤ 1 dB	Derived (measured lesson translated to 106 GBd by model sweep)	Medium
Microbump ≤ 25 fF	Derived (droop scan) — but at an **assumed** R_{eff} = 50 Ω node impedance	Medium
$B_n = 1.5 \times f_{3\text{dB}}$	Convention (design-bounding, §2.3)	Medium — pending measured integrated noise spectra to ≥ 90 GHz
End reflectance ≤ −24 dB	Required (enforced, §3.3)	Medium — manufacturing feasibility judged achievable
PD capacitance 30–40 fF	Assumed (**unmeasured**)	**Low**
MRM EO bandwidth 40–80 GHz bracket	Assumed (no 106 GBd data)	**Low**
MRM peaking-free response	Assumed (**unverified** — the one impairment that could reverse the no-FIR choice)	**Low**
RJ 141 fs rms (analog CDR)	Assumed target (SOA edge; fallback +0.19 dB, §8)	**Low**
+1.34 dB typical closure margin	Model result	Medium — inherits every line above

2. Methodology

2.1 OMA-domain budget framework

The budget is built per DWDM channel in the optical-modulation-amplitude (OMA) domain, between TP2 (transmitter output) and TP3 (receiver input):

$$\mathrm{OMA}_{\mathrm{TX}} - \mathrm{IL}_{\mathrm{link}} \geq \mathrm{OMA}_{\mathrm{floor}} + \sum_i P_i \quad \Longrightarrow \quad \mathrm{margin} = (\mathrm{OMA}_{\mathrm{TX}} - \mathrm{IL}) - (\mathrm{OMA}_{\mathrm{floor}} + \textstyle\sum_i P_i)$$

where $\mathrm{OMA}_{\mathrm{floor}}$ is the analytic amplifier-noise-limited sensitivity (§2.3) and P_i are the power penalties (§2.6). OMA rather than average power is used because NRZ decisions ride on the eye amplitude. Average-power bookkeeping would drag the extinction-ratio penalty $\frac{ER+1}{ER-1}$ into every line, whereas in the OMA domain ER only enters through signal-dependent noise (shot, RIN, MPI) on the individual rails.

TDEC accounting choice. Transmitter eye closure is booked **inside the ISI+EQ line** of the penalty stack (the pulse-response simulation contains the Tx filter), *not* as a separate TDEC subtraction — this avoids double counting. The one exception is the GEN1 closure against the *spec-minimum* transmitter (§3.2), where the spec's own TDEC-coupled OMA floor is exercised and TDEC is subtracted explicitly while the ISI line uses only the reference-receiver filtering.

2.2 BER targets and Personick Q

$\mathrm{BER} = \frac{1}{2}\,\mathrm{erfc}(Q/\sqrt{2})$:

BER	Q	Role
2.4×10^{-4}	3.49	OCI MSA compliance threshold (TDEC, SRS, RxSens)
10^{-6}	4.75	MSA BER-floor requirement
10^{-12}	**7.035**	**Internal design target used throughout**
10^{-15}	7.94	Deep-margin studies

In an amplifier-noise-limited receiver the move from $Q = 3.49$ to $Q = 7.035$ costs $10\log_{10}(7.035/3.49) \approx 3.0$ dB of OMA. At 10^{-12} pre-FEC, the KP4 FEC carries no traffic-dependent burden — it is pure margin. Penalties that scale with Q^2 (RIN, §2.6) grow disproportionately at this target: this is why the internal budget cannot simply re-use spec-threshold penalty values.

2.3 Analytic sensitivity floor from measured TIA data

The floor is calibrated from characterization tables, not assumed:

1. **Input-referral.** For each TIA setting, input-referred noise $i_n = v_{n,\mathrm{out}}/Z_{T,\mathrm{peak}}$, with $v_{n,\mathrm{out}}$ from `TT_Tia_Noise.csv` and $Z_{T,\mathrm{peak}}$ the **peak single-ended p-leg** transimpedance from the TF table. The single-ended convention is the conservative reading of the model (which injects its noise target on the p-leg while the signal sees the differential gain). It is kept for both generations for comparability.

2. **Noise integration bandwidth.** For the signal-dependent noise terms (shot, RIN, dark current): $B_n = 1.5 \times f_{3\mathrm{dB}}$, both generations — the single-pole noise-equivalent-bandwidth factor ($\pi/2$, rounded down), applied as a deliberately conservative convention. The signal-TF Personick integral ($\int |H/H_0|^2 df$ — 28.2 GHz for this design point, or 1.10× $f_{3\mathrm{dB}}$ for an ideal Butterworth-2) is **not** used: it understates a real receiver's noise bandwidth, because physical noise densities extend well beyond the circuit's 3 dB point (the measured family's noise is f^2-shaped, §4.3) — noise must be integrated beyond the bandwidth of the circuit. GEN1 design point: $B_n = 1.5 \times 29.3 = 43.9$ GHz.
3. **Floor.** $\mathrm{OMA}_{\mathrm{sens}} = 2\,Q\,i_n/R.$

Worked GEN1 example (setting `12211111`, the lowest-noise setting with $f_{3\mathrm{dB}} \geq 0.55\times$ baud — 55 of 152 settings meet that gate): $v_n = 5.28$ mV, $Z_T = 1664$ V/A (64.4 dBΩ SE, 70.7 dBΩ differential) → $i_n = 3.17\ \mu$A rms, $B_n = 43.9$ GHz (1.5×29.3), $R = 0.876$ A/W (the `pd` column of the same tables). Then

$$\mathrm{OMA}_{\mathrm{sens}} = \frac{2 \times 7.035 \times 3.17\ \mu\mathrm{A}}{0.876\ \mathrm{A/W}} = 50.9\ \mu\mathrm{W} = -\mathbf{12.93\ dBm}.$$

(Script `01`. At the spec's $Q = 3.49$ the same receiver floors at −15.97 dBm, comfortably inside the −6.2 dBm SRS limit.)

Three noise-bandwidth accountings, compared. To keep the modeling choice explicit (it is the single most consequential convention in the budget — register row 23):

Accounting	GEN1 (29.3 GHz design pt)	GEN2 (58 GHz class)	Standing
Signal-TF shape integral ($\int \lvert H/H_0\rvert^2 df$)	28.2 GHz (0.96×)	64 GHz (1.10×)	Rejected for B_n — a clean-model integral that understates a physical part's noise bandwidth
Measured noise reality	f^2-dominated: density keeps **rising** past $f_{3\mathrm{dB}}$ (§4.3)	**unmeasured** above ~60 GHz	The reason the shape integral is rejected; the eventual ground truth
$\mathbf{1.5} \times \mathbf{f}_{\mathbf{3}\mathrm{dB}}$ (adopted)	43.9 GHz	87 GHz	**Design-bounding convention, pending measured integrated noise spectra to ≥ 90 GHz.** Not a physical fact; costs ~0.33 dB per stack vs the shape integral (§3.4, row 23)

When candidate-TIA noise spectra measured to ≥ $1.5 \times f_{3\mathrm{dB}}$ exist, the measured integral replaces the convention and the ~0.33 dB of deliberate conservatism is either recovered or confirmed as real.

2.4 Pulse-response framework (ISI, CD, jitter share one engine)

A single 1-UI rectangular pulse is propagated through the cascaded frequency response on a 32×-oversampled grid ($N = 8192$):

- **Tx:** two identical cascaded real poles fitted to the specified 20–80 % transition time via the analytic 2-pole step response $s(t) = 1 - (1 + t/\tau)e^{-t/\tau}$ ($t_{20\text{–}80} \approx 3.36\,\tau$). GEN1's spec-referenced Tx uses the 26.5625 GHz 4th-order Bessel-Thomson TDEC reference filter instead.
- **TIA:** measured differential TF interpolated onto the grid, magnitude flattened below 2 GHz with bulk-delay phase preserved — this encodes the **DC-restoration assumption** (a baseline-wander loop handles the AC-coupling droop — §5 turns this into an explicit LF-cutoff requirement). Hypothetical TIAs use 2nd-order Butterworth responses.
- **Microbump (GEN2):** the EIC is bumped directly onto the PIC — the driver drives the modulator (and the PD feeds the TIA) straight through the bump, unterminated. The bump capacitance is booked as a single pole $f_p = 1/(2\pi R_{\mathrm{eff}} C)$ at an assumed effective node impedance R_{eff} (placeholder 50 Ω pending the actual driver R_{out} / TIA R_{in}): 25 fF → 127 GHz. Conservative when the real node impedance is lower, as direct drive intends.
- **ISI metric:** peak distortion on UI-spaced taps around the cursor with the sampling phase optimized for maximum eye: $P_{\mathrm{ISI}} = -10\log_{10}\left[(h_0 - \sum |h_{i\neq 0}|)/h_0\right]$ (optical dB).
- **CD:** quadratic spectral phase $\exp(j\pi D L \lambda^2 f^2/c)$ at the spec corners ±1.7/−0.9 ps/nm. The penalty is the *delta* of the peak-distortion metric.

2.5 Equalization modeling

- **Tx FIR (slice DAC).** Three parallel limiting driver slices (pre/main/post) with 1-UI delays summing at the output node. Because each slice is two-level and the data is NRZ, the summed waveform is exactly a linear FIR on the data — slice nonlinearity does not corrupt tap accuracy. The peak-swing constraint $\sum_i |w_i| = 1$ means de-emphasis costs OMA: the optimizer charges $-10\log_{10}(\sum_i w_i)$ against every candidate tap set (grid search, step 0.02–0.04).
- **Rx CTLE.** One zero, two coincident poles, unit DC gain: $H(s) = (1 + s/\omega_z)/(1 + s/\omega_p)^2$. Each candidate is charged its **noise enhancement**, the rms gain over the TIA-shaped noise: $\eta = \sqrt{\int |H_{TIA} H_c|^2 df / \int |H_{TIA}|^2 df}$, and the sweep minimizes $P_{\mathrm{ISI}} + 10\log_{10}\eta$. Joint FIR+CTLE cells nest the two searches.
- **DFE:** excluded by architecture (GEN2 is CTLE-only). GEN1's budget assumes the standard DSP-SerDes DFE removes post-cursors (§3.1).

2.6 Penalty formulas and inputs

Penalty	Model	Inputs
ER / shot	Q-solve with level-dependent shot on both rails: $\sigma_L^2 = i_n^2 + 2qRP_L B_n$	ER 3.5 dB (GEN1 spec min) / 4.5 dB (GEN2 target)
RIN	Q-solve with $\sigma_{RIN,L} = RP_L\sqrt{\mathrm{RIN}\cdot B_n}$, penalty scales as Q^2	RIN$_{\mathrm{OMA}}$ −138 dB/Hz @ 21.4 dB ORL. **RIN-floor check:** $Q_{\max} = [\sqrt{\mathrm{RIN}\,B_n}\,(P_1 + P_0)/\mathrm{OMA}\,]^{-1} = 14.5$ → BER floor $\sim 10^{-47}$, no floor issue
MPI	Bhatt/King discounted upper bound: $P = -10\log_{10}(1-x)$, $x = 4DS\frac{E}{E-1}$, $S = \sqrt{R_t R_r} + n\sqrt{R_t R_c} + n\sqrt{R_r R_c} + \frac{n(n-1)}{2}R_c$	$D = 0.5$. Ends −24 dB for both generations (shared product line, §3.3; spec allows −19 dB → 0.51 dB), 4 connectors −35 dB. GEN1 (ER 3.5): 0.24 dB. GEN2 (ER 4.5): 0.205 dB
ISI + EQ	§2.4/§2.5 net of noise enhancement and de-emphasis	per scenario

Penalty	Model	Inputs
CD	pulse-sim delta at ±1.7/−0.9 ps/nm	GEN1 0.005 → book 0.01; GEN2 sim 0.015 → book 0.04 (scales ≈ baud²)
Jitter	dual-Dirac: $TJ = DJ + 2Q\sigma_{RJ}$; penalty from the equalized eye re-evaluated at $\pm TJ/2$; a closed eye at the offset ⇒ scenario does not close (no finite penalty is booked)	GEN1: RJ 0.010 UI rms, DJ 0.10 UI → TJ 0.241 UI (4.5 ps) → 0.61 dB. GEN2: RJ 0.015 UI (141 fs), DJ 0.14 UI → TJ 0.351 UI (3.30 ps) → 0.95 dB
Inter-channel crosstalk	$-10\log_{10}(1-2\varepsilon)$, $\varepsilon = n_{adj}10^{-\text{iso}/10}10^{\Delta\text{OMA}/10}$	MRR demux isolation 20 dB (assumed), 2 neighbors, +3 dB aggressor ΔOMA → 0.36 dB
Threshold offset	$10\log_{10}(1+2\delta)$	$\delta = 2.5\%$ of swing → 0.21 dB
Dark current	$10\log_{10}\sqrt{1+2qI_{DK}B_n/i_n^2}$	1 µA → 0.003 dB

3. GEN1 budget — 53.125 GBd NRZ (script `02`)

3.1 Penalty stack

Receiver: design-point TIA (§2.3), Tx = BT4 26.5625 GHz reference, DFE removes post-cursors. Unequalized peak-distortion ISI is 2.15 dB (cursor 0.809, pre 0.111, post 0.205); the DFE leaves the pre-cursor-only residual.

Line	dB	Derivation
ER / signal-dependent shot	0.18	Q-solve at ER 3.5 dB, B_n 43.9 GHz (1.5× $f_{3\text{dB}}$)
RIN (−138 dB/Hz, Q^2-scaled)	0.68	4.5× its value at spec BER (0.15 dB) — the Q^2 effect
MPI (Bhatt UB, $D = 0.5$)	0.24	ends −24 dB (shared GEN1/GEN2 product line, §3.3), 4 × −35 dB connectors, ER 3.5
ISI residual after EQ	0.64	$-10\log_{10}[(0.809-0.111)/0.809]$
Chromatic dispersion	0.01	+1.7 ps/nm worst corner (computed 0.005)
Jitter	0.61	TJ = 0.241 UI, eye at ±TJ/2
Inter-channel crosstalk	0.36	20 dB isolation, +3 dB ΔOMA
Decision threshold offset	0.21	δ = 2.5 %
Dark current	0.00	1 µA
Total	**2.93**	

3.2 Closure

Required OMA at Rx $= -12.93 + 2.93 = \mathbf{-10.00}$ dBm.

Scenario	Tx OMA	−IL	−TDEC	At Rx	Margin
Spec-min Tx, TDEC 1.4 dB	−5.5	2.5	1.4	−9.40	**+0.60 dB**
Spec-min Tx, TDEC 3.4 dB	−3.5	2.5	3.4	−9.40	**+0.60 dB**
Realistic Tx (−3.2 dBm, TDEC ≈ 2 dB)	−3.2	2.5	2.0	−7.70	**+2.30 dB**

The spec's OMA-min law $\max(-5.5, -6.9 + \text{TDEC})$ pins OMA−TDEC at −6.9 dBm for TDEC ≥ 1.4 dB, so both spec corners close identically. The realistic case is grounded in the spec's own stressed-test aggressor point (−3.2 dBm), the ELS power budget, and the ~1.2 V$_{\mathrm{ppd}}$ swing of the `Typ_Txdrv_NL.csv` driver class at ER ≈ 4.5 dB. TDEC is applied unscaled (defined at BER 2.4×10^{-4}; its deterministic content does not scale with Q — noted caveat).

Two independent cross-checks: (i) the spec itself closes by construction with exactly its 0.2 dB MPI allowance (−3.5 − 2.5 = −6.0 dBm vs −6.2 dBm SRS); (ii) the COUPE BIDI simulated waterfall crosses BER 10^{-12} at −11.8 dBm OMA — between our analytic floor (−12.93) and full required OMA (−10.00), consistent with a simulation that includes ISI/jitter but not MPI/crosstalk stresses.

3.3 IEEE Clause 181 cross-check and the MPI finding

Our 2.93 dB stack sits 1.0 dB under Clause 181's 3.9 dB allocation — expected, since the OCI channel is shorter/lower-loss and NRZ needs no PAM4 level-separation penalty. At the spec's −19 dB end reflectances the MPI line would be 0.51 dB, matching Clause 181's 0.5 dB rather than the OCI spec's 0.2 dB allocation: with $S = 0.0304$, the two −19 dB ends contribute 93 % of S (41 % direct pair + 52 % end-connector cross terms), and the worst case ($D = 1$) reaches 1.08 dB. Meeting ~0.2 dB at ER 3.5 dB requires roughly $R_t = R_r \le -24$ dB.

Requirement adopted: GEN1 and GEN2 are the same product line, so the GEN2-derived ≤ -24 dB end-reflectance requirement (§4) is enforced for both generations — the ~0.2 dB MPI line is ensured by the reflectance criterion rather than assumed. The GEN1 budget therefore books MPI at **0.24 dB** ($D = 1$ worst case: 0.49 dB). A part meeting only the spec's −19 dB would add 0.27 dB back to the stack (0.51 dB line).

3.4 Sensitivity to assumptions (vs the +0.60 dB base)

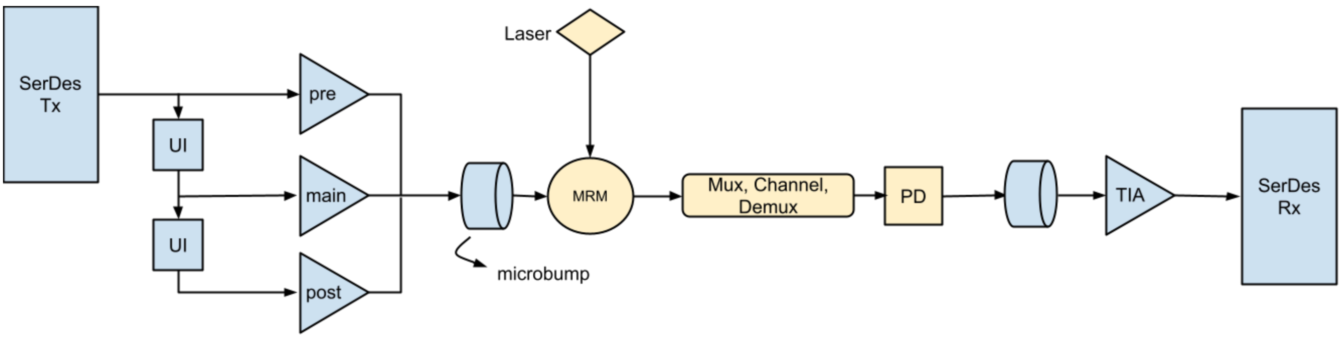
Variation	Effect	New margin
TIA noise 3.17 → 4.26 μA (median usable setting)	floor +1.28 dB	−0.69 dB
TIA noise → 6.75 μA (worst usable)	floor +3.28 dB	−2.68 dB
R 0.876 → 0.80 / 1.00 A/W	floor +0.39 / −0.58 dB	+0.20 / +1.17 dB
MPI: 2 / 6 connectors	0.24 → 0.14 / 0.36 dB	+0.70 / +0.48 dB
MPI: $D = 1$ worst case	0.24 → 0.49 dB	+0.35 dB
MPI: ends at spec −19 dB (shared −24 dB assumption fails)	0.24 → 0.51 dB	+0.33 dB
RIN −138 → −144 dB/Hz (ELS-only)	0.68 → 0.16 dB	+1.13 dB
B_n: signal-TF shape integral (28.2 GHz) instead of 1.5× f_{3dB}	ER/shot+RIN 0.86 → 0.53 dB	+0.93 dB

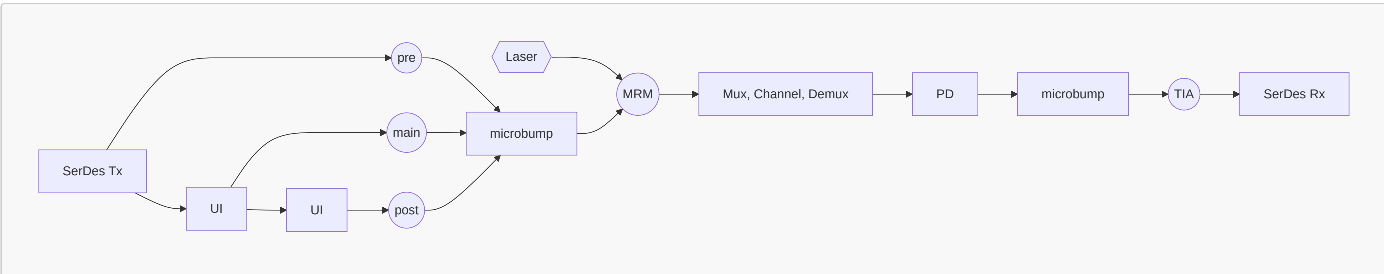
TIA noise dominates the GEN1 budget — which is precisely why the GEN2 exercise (§4–5) is framed as deriving the receiver class the doubled rate requires.

---

4. GEN2 CPO architecture and rate scaling — 106.25 GBd NRZ (scripts `03`, `04`)

4.1 Architecture





Per the block diagram: SerDes Tx feeds a cascaded delay line (two 1-UI / 9.412 ps delays), with the `pre`/`main`/`post` **limiting driver slices** tapped off the undelayed, once-delayed, and twice-delayed nodes respectively and summed at the output node — this is the analog FIR — through an EIC–PIC microbump to the MRM (the laser feeds the MRM directly), mux → 500 m fiber → demux, PD, second microbump, TIA inside the SerDes Rx AFE. **Rx EQ is CTLE only; DFE is not supported.** CPO eliminates the package trace channel (1.61/0.72 dB differential IL at the new Nyquist, plus its reflections) and replaces it with a microbump parasitic. The bump is an unterminated direct connection — extra load capacitance, not a matched interface. Booked as a single 127 GHz pole at 25 fF (§2.4): 0.70 dB droop at 53.1 GHz, costing +0.22 dB of equalized ISI (1.15 vs 0.93 dB); 50 fF would cost +0.71 dB — hence the tightened ≤ 25 fF / ≤ 30 pH microbump requirement. A low driver output impedance shrinks this charge.

4.2 What scales with rate

Quantity	GEN1 → GEN2	Consequence
UI	18.82 → 9.412 ps	all ps-domain tolerances halve (FIR tap spacing ±0.47 ps, slice DCD ≤ 0.47 ps)
Nyquist	26.56 → 53.125 GHz	receiver class must move to 0.5–0.6× of the new baud
Noise bandwidth (1.5 × f_{3dB}, §2.3)	43.9 → 87 GHz	floor rises; shot/RIN integrate over ~2× the bandwidth (ER/shot+RIN 0.86 → 1.16 dB)
Jitter in ps	TJ 4.5 → 3.30 ps at similar UI budget	RJ spec becomes 141 fs rms — analog-CDR state of art
CD	~baud²: 0.01 → 0.04 dB booked	still small at 500 m
Tx transition	10/12/17 ps corners are 1.1–1.8 UI	new corners defined as 0.35/0.45/0.60 UI (3.3/4.2/5.6 ps; 2-pole fits: 105/82/61 GHz poles)

4.3 Receiver noise scaling law (measured-family calibration)

A non-negative least-squares regression of i_n^2 against the white and f^2 noise integrals ($B_1 = \int |\hat{H}|^2 df$, $B_2 = \int f^2 |\hat{H}|^2 df$, including the model's 60 GHz noise-path LPF) across all 152 settings shows the family is f^2**-noise dominated**: single-parameter fits give $R^2 = 0.87$ (f^2-only) versus 0.59 (white-only). Scaling the GEN1 design point to a 58 GHz Butterworth-2 response (white shape-integral $B_1 = 64.4$ GHz, 2.92× GEN1's — these scaling integrals are distinct from the §2.3 noise-integration-bandwidth convention):

- white ($\sqrt{BW}$) scaling: $i_n = 5.42\ \mu$A → floor −10.60 dBm (the optimistic bound);
- f^2 ($BW^{1.5}$) scaling: $i_n = 17\ \mu$A → floor −5.63 dBm (the fit-favored law).

This scaling law — not any pass/fail statement — is the measured data's main contribution to GEN2: it demonstrates that the GEN2 noise line cannot be reached by re-tuning this input stage, and quantifies the gap a new design must close. (For completeness, a full 152-setting re-scan at 106.25 GBd with per-setting CTLE optimization confirms it: no setting qualifies, which is expected and unremarkable — the family was designed for the GEN1 rate. The scan's real yield is the group-delay failure mode in §5.)

4.4 GEN2 floor, stack, and closure (derived TIA class)

Target-class TIA: Butterworth-2 at 58 GHz, $i_n = 4.5\ \mu$A rms total; shot/RIN integrate over $B_n = 1.5 \times 58 = 87$ GHz (§2.3 convention) — in the same range as 100 GBd-class TIAs the team is aware of (roughly 2.5–5 μA at 55–65 GHz), though this bracket is an internal engineering estimate, not a cited publication (see Methodology_Provenance.md). Floor:

$$\mathrm{OMA_{floor}} = \frac{2 \times 7.035 \times 4.5\ \mu\mathrm{A}}{0.876} = 72.3\ \mu\mathrm{W} = -\mathbf{11.41\ dBm}.$$

Penalty stack (typical Tx 0.45 UI, 25 fF bump in chain, optimal CTLE $z = 37$ GHz / $p = 62$ GHz, noise enhancement ×0.99):

Line	GEN1 CPO @ 53 GBd	GEN2 @ 106 GBd	Driver
ER/shot + RIN (Q-solve)	0.86	1.16	B_n 43.9 → 87 GHz (1.5× $f_{3\mathrm{dB}}$)
MPI	0.21	0.21	rate-independent; −24 dB ends, ER 4.5
ISI + EQ net	1.86	1.15	both assume a 0.55×-baud TIA (measured at 53 G, derived class at 106 G)
CD	0.01	0.04	~baud²
Jitter	0.73	0.95	similar UI budget, halved UI
Crosstalk	0.36	0.36	kept
Threshold	0.21	0.21	kept
Total	**4.23**	**4.07**	the real rate cost is in the floor (+1.5 dB) and component specs, not the stack total

Closure at TP2 = −3.5 dBm (TP3 = −6.0 dBm after 2.5 dB IL):

Case	Floor	Stack	Required OMA	Margin
Fast Tx 0.35 UI	−11.41	3.74	−7.67	**+1.67 dB**
Typical Tx 0.45 UI (baseline)	−11.41	4.07	−7.34	**+1.34 dB**
Max Tx 0.60 UI	−11.41	4.62	−6.79	**+0.79 dB**
White-scaled existing input stage (5.4 μA)	−10.60	4.04	−6.57	+0.57 dB
f^2-scaled (17 μA)	−5.63	3.91	−1.73	−4.27 dB

GEN1 reference at the same Tx OMA: floor −12.93, stack 4.23, margin +2.70 dB. Inverting the stack for the noise requirement: +2 dB margin needs $i_n \le 4.03\ \mu$A (13.6 pA/√Hz average over the 87 GHz B_n) with the pre-microbump ISI accounting, or 3.83 μA with the bump charged — the spec line is written as ≤ 4.0 μA with the margin target read as +1.9–2.0 dB. The 4.5 μA target class itself closes at +1.34 dB typical. Tx OMA at −1.5 dBm is the system-level relief valve: +2 dB to every case.

Supporting GEN2 numbers: TDEC proxy through the 53.125 GHz BT4 reference receiver = 0.77/1.11/1.77 dB (fast/typ/max) → spec TDEC ≤ 1.8 dB; slice-DAC FIR analysis shows 3 taps remain sufficient ($h_{+2} \le 0.01$; a 4th tap buys 0.00 dB even on the worst measured chain), with tap spacing 9.412 ± 0.47 ps and weight step ≤ 0.02.

---

5. Derived GEN2 TIA requirements (script `05`) — the centerpiece

The budget exists to produce this table. Every line is derived from the §2 framework; the measured family enters as calibration (noise scaling law, §4.3) and as the source of one lesson learned (group delay, below).

Parameter	Requirement	Derivation
Bandwidth ($f_{3\mathrm{dB}}$, differential)	**50–64 GHz window** (53–58 GHz target; 0.50–0.55× baud)	Sweep of total sensitivity (white-scaled floor + ISI+EQ + RIN at $B_n = 1.5 f_{3\mathrm{dB}}$) vs $f_{3\mathrm{dB}}$ is flat at −8.3/−8.4 dBm across 45–64 GHz (−8.38 at 50–53, −8.34 at 58, −8.26 at 64, −8.15 at 70): below the window ISI grows faster than noise saved; above it noise (floor and RIN band) grows ~0.1–0.2 dB/6 GHz for < 0.3 dB ISI benefit. A window spec, not a point spec. Assumed source: PD 30–40 fF + 25 fF bump + ~10 fF pad ≈ 65–75 fF
Input-referred noise, total	**≤ 4.0 μA rms** over $B_n = 1.5 \times f_{3\mathrm{dB}}$ (87 GHz at the 58 GHz	Q-solve inversion of the full stack (§4.4) at $B_n = 87$ GHz: 4.0 μA → ≈ +2 dB margin (pre-bump accounting; 3.83 μA with the bump charged); 6.5 μA → zero margin. The 4.5 μA target class closes at

Parameter	Requirement	Derivation
	reference); 6.5 µA absolute fail line	+1.34 dB typical
Noise density mask	≤ 14 pA/√Hz band average; ≤ 16 pA/√Hz spot 1–53 GHz; measure the density to ≥ $1.5 \times f_{3\mathrm{dB}}$ (≈ 90 GHz) before integrating	4.0 µA ÷ √(87 GHz) = 13.6 pA/√Hz. Shape check: an f^2-shaped input density at equal total rms sees **≤ 0 dB** extra post-CTLE penalty vs white (−0.98 to −0.65 dB for corners 15–40 GHz) because the mild optimal CTLE's poles sit where the f^2 noise lives. The f^2 danger is rms inflation with bandwidth (the §4.3 scaling law), so the spec binds full-band rms + a spot ceiling instead of a CTLE allowance
Magnitude peaking	**≤ 1.0 dB** over DC–53 GHz, monotonic rolloff above the peak	Variable-Q 2nd-order sweep at fixed 58 GHz $f_{3\mathrm{dB}}$: 1.25 dB peaking ($Q = 1.0$) costs +0.11 dB over the 1.15 dB ISI+EQ line; 2.4 dB costs +0.33; 6.3 dB → 4.27 dB (unusable)
Group-delay ripple	**≤ 3 ps p-p over 2–40 GHz** (0.32 UI)	Lesson from characterization data: the widest measured settings are flat to 0.2 dB in magnitude yet carry 12.5 ps of GD ripple → $h_{-1} \approx 0.48$ pre-cursor that no CTLE can remove (CTLE shapes magnitude; pre-cursor needs phase). 3–4 ps keeps $h_{-1} \le 0.12$ and ISI+EQ within +0.1 dB. The GEN1 design point's 7.15 ps is benign at 53 GBd (0.38 UI) — the ripple budget halves in UI terms with the rate. Phase data is a first-class deliverable
Transimpedance gain Z_T	**≥ 57 dBΩ** (≥ 700 Ω) differential at max gain	Eye current at sensitivity $= 2Q i_n = 63\ \mu$App; a 10–15 mV$_{\mathrm{ppd}}$ slicer needs only 44–48 dBΩ, but keeping TIA noise ≥ 3× an assumed ≤ 1 mV rms downstream (CTLE+slicer) noise sets 56.5 → 57 dBΩ. Measured family spans 58.7–87.1 dBΩ — met by existing gain classes
Overload / dynamic range	No BER degradation 160–700 µApp; gain adjustment ≥ 14 dB in ≤ 1 dB steps	Required-OMA floor −7.34 dBm → 162 µApp; Rx OMA max −1 dBm → 696 µApp (12.7 dB electrical). Output must stay inside the CTLE linear range — hence AGC steps
DC handling	Cancellation ≥ 750 µA	Max average power ≈ −0.8 dBm (OMA −1 dBm at ER 4.5) × 0.876 A/W = 731 µA
LF cutoff	**≤ 1 MHz**, baseline-wander penalty ≤ 0.05 dB at 72-bit CID	Droop ≈ $2\pi f_{LF} N\, UI$: 1.34 MHz gives 0.05 dB at $N = 72$. Also closes the loop on the §2.4 DC-restoration assumption

Rx OMA max governing case (reviewed). `RX_OMA_MAX_DBM = -1.0` (script `05`) is a judgment call, not derived from a documented Tx-OMA/link-IL corner. An adversarial review raised an alternative +1.79 dBm / 1.13 mA${\mathrm{pp}} max − power case (20.8 dB electrical dynamic range) from an external "parallel budget"; if a real deployed variant of this shared TIA macro combines a highe
3.5 to − 1.5 dBm/ 2.5–3 dB corners, that variant's \mathrm{OMA{Tx,max}} - \mathrm{IL_{min}}$ would be the true ceiling and the 696 µApp / 750 µA / 14 dB lines below would need to be re-derived against it. No such variant is documented in this repo's materials, so −1 dBm is retained as the governing case for now; this is a deliberate decision on unchanged evidence, not an unexamined assumption, and should be revisited if a higher-power variant of the product line is confirmed to share this TIA macro.

Hard gates vs. targets. The table mixes two kinds of lines, and the distinction matters at design review:

- **Hard gates** (a part failing any of these does not close the link or violates the spec; these are the `verify_tia()` pass/fail lines): $f_{3\mathrm{dB}}$ inside 50–64 GHz; $i_n \le 4.0\ \mu$A over $B_n = 1.5 f_{3\mathrm{dB}}$ (absolute fail at 6.5 µA); peaking ≤ 1.0 dB; GD ripple ≤ 3 ps; ISI+EQ ≤ 1.15 dB typ (≤ 1.70 at 0.60 UI); jitter penalty ≤ 1.0 dB; end-to-end margin ≥ +1.5 dB.
- **Targets** (preferred design points with margin behind them, not pass/fail): the 53–58 GHz bandwidth center; the 4.5 µA target class (closes at +1.34 dB, i.e. a 4.5 µA part is *usable*, just below the +2 dB spec intent); Tx 0.45 UI corner (0.60 UI still closes at +0.79 dB); Tx OMA −3.5 dBm baseline (−1.5 dBm is the relief valve).

Verification recipe (implemented as `verify_tia()` in script `05`; runs on the same data format as the characterization tables):

1. Measure differential TF (100 kHz–≥90 GHz) and output-noise spectrum to ≥ $1.5 \times f_{3\mathrm{dB}}$ (≈ 90 GHz); input-refer (i_n = integrated output rms ÷ peak single-ended gain); require $i_n \le 4.0\ \mu$A and the density mask.
2. Static response: $f_{3\mathrm{dB}}$ 50–64 GHz; peaking ≤ 1.0 dB with monotonic rolloff; GD ripple ≤ 3 ps (2–40 GHz).
3. Analytic floor $2Qi_n/R \le -11.9$ dBm.
4. Pulse-response flow: 2-pole Tx at 0.45 UI × 25 fF bump × measured TF; optimize CTLE (zero 10–40 GHz, poles 45–95 GHz); require ISI+EQ ≤ 1.15 dB (≤ 1.70 dB re-run at 0.60 UI).
5. Jitter: with TJ = 0.351 UI dual-Dirac, eye at ±TJ/2 open with penalty ≤ 1.0 dB.
6. End-to-end: floor + full stack ≤ −7.5 dBm required OMA, i.e. margin ≥ +1.5 dB at Tx OMA −3.5 dBm.

Buildability. 4.0 µA total (integrated to 1.5× $f_{3\mathrm{dB}}$) with a 65–75 fF input node is state of the art; it sits in the same 2.5–5 µA / 55–65 GHz range the team associates with 100 GBd-class TIAs (SiGe or advanced FinFET with T-coil input peaking), but that range is an unreferenced internal estimate rather than a cited data point (see Methodology_Provenance.md). The gain line is already met by existing gain classes; the requirements table defines a new input-stage design class, with the noise density mask and the phase-quality line as the two genuinely new asks.

---

6. Tx/driver requirements and device trade-offs (script `06`)

Real device brackets evaluated as special cases of the GEN2 budget: TIA A (3–4 µA at 50 GHz), TIA B (4–5 µA at 60 GHz), and a no-FIR transmitter whose driver reaches 60 GHz. TIAs modeled as clean Butterworth-2 responses **assumed to meet the §5 peaking/GD lines** — the load-bearing assumption; gate any selection on §5 step 2.

6.1 Scenario matrix (margin at Tx OMA −3.5 dBm, BER 10^{-12})

Tx case	A @ 3 µA/50 G	A @ 4 µA/50 G	B @ 4 µA/60 G	B @ 5 µA/60 G
FIR3, typ driver (0.45 UI)	+2.81	+1.63	+1.98	+1.06
FIR3, slow driver (0.60 UI)	+2.22	+1.04	+1.40	+0.48

Tx case	A @ 3 µA/50 G	A @ 4 µA/50 G	B @ 4 µA/60 G	B @ 5 µA/60 G
no-FIR, 60 G driver + MRM 80 GHz	+2.80	+1.62	+1.83	+0.92
no-FIR, 60 G driver + MRM 60 GHz	+2.50	+1.32	**+1.63**	+0.72
no-FIR, 60 G driver + MRM 50 GHz	+2.25	+1.07	+1.42	+0.50
no-FIR, 60 G driver + MRM 40 GHz	+1.95	+0.77	+1.07	**+0.16**

Every cell closes — the trade is margin, risk, and complexity, not feasibility. The worst-everything cell (+0.16 dB) is now thin, though: it leans on the Tx OMA relief valve (§8) if any further assumption slips.

6.2 What the FIR actually buys: nothing, on this channel

Isolation experiment on an identical 60 + 60 GHz channel (TIA B @ 4 µA): the joint FIR+CTLE optimizer **converges to zero tap weight** — ISI+EQ = 1.66 dB with or without the FIR. The channel taps explain it: $h_{-1} = 0.09$, $h_{+1} = 0.22$, nothing beyond — a mild post-cursor the CTLE removes at ≈ 1.04× noise cost, while the peak-constrained slice FIR must pay ~1:1 in de-emphasis OMA for the same eye. The FIR's only net contribution is its slice-DCD jitter charge: DJ 0.14 vs 0.11 UI → jitter 0.85 vs 0.58 dB — **no-FIR wins by 0.27 dB**, before counting the ±0.47 ps delay-line accuracy and slice matching it eliminates. Even on the 40 GHz-MRM channel, re-enabling the FIR buys zero ISI (the optimizer again picks zero taps): a slow MRM argues for Tx OMA headroom, not for the FIR. The 60 GHz→edge mapping (20–80 % ≈ $0.345/f_p$ per pole; the 60+60 composite measures 5.88 ps = 0.62 UI) and the MRM sweep (80→40 GHz costs ~0.8 dB, cleanly recoverable post-cursor) bound the risk.

6.3 TIA A vs B and the recommendation

At the 4 µA overlap, B beats A by ~0.35 dB in every Tx case (+1.98 vs +1.63 typ): the 50 GHz TIA's extra ISI (+0.54 dB, needing 81–93 GHz CTLE poles at 1.11–1.17× noise) outweighs its noise-bandwidth savings. The crossover sits at $i_n \approx 3.6\ \mu$A — below that, A's floor advantage dominates (A @ 3 µA is the best cell, +2.81 dB).

Recommendation: no-FIR 60 GHz driver + TIA B as the preferred baseline; retain a single post-cursor tap as a design option pending measured MRM EO response — and switch to option A only if it demonstrably lands below ~3.5 µA. This is a baseline choice, not a closed conclusion: it holds on the modeled clean channel and is explicitly conditioned on (i) the **phase-quality gate** — verify §5 peaking/GD on the real silicon before trusting any cell; (ii) the **MRM peaking check** — real MRM detuning peaking/overshoot is the one thing a Tx FIR fixes that a CTLE cannot, and strong measured peaking would reverse the no-FIR call in favor of the retained post tap; (iii) **large-swing edge rate** — if the driver's 60 GHz is a small-signal number, use the measured large-swing edge (each +0.15 UI of transition costs ~0.5 dB).

---

7. Assumptions register

#	Assumption	Value	Source	Margin sensitivity
1	Personick Q at 10^{-12}	7.035	derived (erfc)	definitional
2	PD responsivity R	0.876 A/W	measured (pd column, TIA TF tables)	±0.08 A/W → ∓0.4–0.6 dB floor
3	TIA input-referral convention	SE p-leg peak gain	derived from model structure (conservative)	differential convention would improve floors ~0.3 dB
4	ER	3.5 dB (GEN1 spec min) / 4.5 dB (GEN2 target)	spec / assumed	enters shot/RIN/MPI lines
5	RIN$_{\mathrm{OMA}}$	−138 dB/Hz	spec (@ 21.4 dB ORL)	−144 (ELS-only): GEN1 RIN line 0.68 → 0.16 dB (margin +1.13 at spec-min)
6	GEN1 jitter	RJ 0.010 UI rms, DJ 0.10 UI	assumed (DSP-SerDes class)	jitter line 0.61 dB
7	GEN2 jitter	RJ 0.015 UI = 141 fs, DJ 0.14 UI (incl. 0.05 slice DCD)	assumed; 113 fs judged beyond analog-CDR SOA	jitter line 0.95 dB; no-FIR credits DJ→0.11 (worth 0.27 dB)
8	Crosstalk isolation	20 dB adjacent, 2 neighbors, +3 dB ΔOMA	assumed (MRR demux)	0.36 dB line; 17 dB iso ≈ +0.4 dB
9	Threshold offset	2.5 % of swing	assumed (offset cal)	0.21 dB line
10	Dark current	1 µA	assumed worst case	negligible
11	MPI discount D	0.5	Bhatt/King	$D = 1$: GEN1 MPI 0.24→0.49 dB
12	End reflectance	≤ −24 dB both generations (shared product line; OCI spec only requires −19 dB); connectors 4 × −35 dB	**enforced requirement** (§3.3/§4) — the ~0.2 dB MPI line is ensured by this criterion, not assumed	MPI 0.24 (GEN1) / 0.205 dB (GEN2); a part missing it at −19 dB books 0.51 dB (−0.27 dB margin)
13	Link IL	2.5 dB (500 m)	spec	dB-for-dB
14	Microbump	≤ 25 fF, ≤ 30 pH; booked as a single 127 GHz pole (unterminated direct drive, §2.4)	derived limit (droop scan); effective node impedance **assumed** pending driver R_{out} / TIA R_{in}	50 fF → +0.5 dB ISI; a low driver output impedance shrinks the +0.22 dB charge
15	GEN2 Tx corners	0.35/0.45/0.60 UI 20–80 %	assumed corners	0.45→0.60 UI costs 0.55 dB

#	Assumption	Value	Source	Margin sensitivity
16	MRM EO bandwidth	40–80 GHz sweep; no 106 GBd MRM data in repos	user-provided bracket / assumed	80→40 GHz ≈ −0.8 dB (no-FIR case)
17	PD capacitance	30–40 fF	assumed (100G-class waveguide PD; **unmeasured**)	no dB line of its own in the penalty stack — enters the budget only through whether the §5 TIA class ($f_{3\mathrm{dB}}$ /noise at a 65–75 fF input node) is buildable; 60 fF would strand the 4.0 μA line
18	Slicer sensitivity	10–15 mV$_{\mathrm{ppd}}$; downstream noise ≤ 1 mV rms	assumed (lecture-4 class)	sets $Z_T \geq 57$ dBΩ line
19	CID run length	72 bits, 0.05 dB droop	assumed (scrambled NRZ)	sets LF cutoff ≤ 1.34 → 1 MHz
20	Tx OMA baseline	−3.5 dBm (GEN2)	spec-min at TDEC 3.4	−1.5 dBm relief adds +2 dB everywhere
21	TDEC treatment	unscaled from 2.4e-4 to 1e-12	stated caveat	conservative-neutral
22	Target-class TIA noise	4.5 μA total ($B_n = 87$ GHz for shot/RIN)	assumed; internal 2.5–5 μA / 55–65 GHz estimate for the 100 GBd class, **not a cited source** (§5, Methodology_Provenance.md)	±0.5 μA ≈ ∓0.9–1.0 dB floor
23	Noise integration bandwidth	$B_n = 1.5 \times f_{3\mathrm{dB}}$, both generations (GEN1 43.9, GEN2 87 GHz)	**convention** — single-pole NEB factor $\pi/2$ rounded; real noise extends beyond $f_{3\mathrm{dB}}$ (§4.3 f^2 finding), so shape integrals (0.96–1.10× $f_{3\mathrm{dB}}$) understate it	vs shape-integral accounting: GEN1 stack +0.33 dB, GEN2 stack +0.34 dB

8. Risks and open items

GEN2 margin sensitivity, ranked (impact on the +1.34 dB typical margin; from the §7 register sensitivity column and script sweeps — a tornado chart in table form):

Lever	Swing	Margin impact
Tx OMA relief valve	−3.5 → −1.5 dBm	**+2.0 dB**
TIA noise	±0.5 μA around 4.5 μA	**∓0.9–1.0 dB**
MRM EO bandwidth	80 → 40 GHz	−0.8 dB
Tx transition corner	0.45 → 0.60 UI	−0.55 dB
Microbump capacitance	25 → 50 fF	−0.5 dB
Crosstalk isolation	20 → 17 dB	≈ −0.4 dB
B_n convention	1.5× $f_{3\mathrm{dB}}$ → shape integral	+0.34 dB
End reflectance	−24 → −19 dB	−0.31 dB
CDR RJ fallback	141 → 169 fs	−0.19 dB
PD capacitance	30–40 → 60 fF	no direct dB line — **feasibility gate** on the §5 TIA class (strands the 4.0 μA requirement)

TIA noise and the Tx OMA valve dominate; PD capacitance is not a margin lever but a buildability gate. This ranking is where validation effort should go first.

1. **MRM peaking and nonlinearity are unmodeled.** The MRM enters as a clean pole. Real detuned-MRM peaking/overshoot is the one impairment a Tx FIR fixes that a CTLE cannot — it could reverse the no-FIR recommendation (keep one post tap as insurance if MRM data shows peaking). 106 GBd MRM EO-bandwidth data does not exist in the repos.
2. **PD capacitance is unmeasured.** The 30–40 fF input assumption underpins the TIA bandwidth/noise feasibility argument; a 60 fF PD would make the 4.0 μA line materially harder.
3. **TIA phase data is required.** The GD-ripple line (≤ 3 ps) came from characterization data showing 12.5 ps ripple with flat magnitude; candidate-TIA phase (or time-domain pulse) measurements to 80 GHz must be part of the vendor deliverable, or §5 step 4 cannot be run.
4. **Analog-CDR RJ at 141 fs rms** is at the edge of published art at 106 GBd; a 0.018 UI (169 fs) fallback spec, run through the same typ-Tx/target-TIA jitter chain as §4.4, costs an incremental **+0.19 dB** over the 141 fs baseline (script `04`, re-evaluated at 8× finer time-grid resolution to avoid a sample-quantization artifact — see Methodology_Provenance.md) and should be pre-negotiated in the jitter budget.
5. **Tx OMA relief valve.** Raising launch OMA from −3.5 to −1.5 dBm buys 2 dB across every scenario — the single most effective system-level knob if any component assumption slips (ELS power and MRM insertion loss permitting).
6. **Microbump R_{eff} is a placeholder.** The 127 GHz bump pole assumes a 50 Ω effective node impedance (§2.4), but the real driver→MRM and PD→TIA nodes are unterminated direct-drive interfaces that will not behave as a simple 50 Ω lump. The booked +0.22 dB ISI charge is conservative if the true node impedance is lower and optimistic if higher; extracting driver R_{out} / TIA R_{in} from circuit design and re-running the droop scan is required before the ≤ 25 fF line can be treated as final.

Next experiments — the kill-or-confirm set. The architecture claim reduces to a finite, measurable list: *106.25 GBd NRZ CPO is feasible provided the TIA achieves the derived noise/BW/phase class, PD capacitance stays in the assumed range, the MRM introduces no problematic peaking, and the clock/CDR approaches the RJ* *target.* The remaining work is predominantly characterization, not further model iteration — these four experiments retire the Low-confidence rows of the §1 status table:

1. **Measure PD capacitance and the MRM EO response (S21 magnitude + phase, or impulse response) at 106 GBd-relevant bandwidths.** Retires the two unbounded Rx and Tx assumptions at once: the 30–40 fF input-node term behind the §5 TIA class, and the 40–80 GHz MRM bracket plus the peaking question that gates the no-FIR decision (§6.3).
2. **Measure candidate-TIA magnitude, phase, and output-noise spectrum to ≥ 90 GHz** ($\geq 1.5 \times f_{3\text{dB}}$). Feeds `verify_tia()` directly (§5 recipe), replaces the B_n convention with the measured noise integral (§2.3), and tests the GD-ripple line on real silicon.
3. **Characterize the actual clock/CDR RJ** against the 141 fs rms target (and the 169 fs / 0.018 UI fallback, item 4 above). The jitter line is currently the only hard-gate input with no measured-part backing at all — items 1–2 at least start from repo characterization data; RJ is a pure forward assumption.
4. **Run the complete measured chain end-to-end** — EIC driver → bump → MRM → fiber → PD → bump → TIA — through the §2.4 pulse engine with all-measured blocks, and compare against the stack's model-based ISI+EQ, jitter, and TDEC lines. This is the closure test that converts the +1.34 dB from a model result into a demonstrated one.

If all four land inside the assumptions register, the budget has done its job: it has converted "can 106G CPO NRZ work?" into a finite set of measurable component requirements, per the §1 bottom line.

---

9. Appendix A: reproduction guide

Script-to-result map

Result	Script	Canvas
152-setting survey; GEN1 design point (3.17 μA / 29.3 GHz); floor −12.93 dBm	`01_tia_survey.py`	bottom-up §1
GEN1 stack 2.93 dB; required −10.00 dBm; +0.60 / +2.30 dB; sensitivity table	`02_bottom_up_budget_53g.py`	bottom-up §2–5
53 GBd CPO: Tx corners, EQ scenarios (CTLE-only 1.86 dB), stack 4.23, +2.70 dB; microbump/package	`03_cpo_gen2_53g.py`	GEN2 spec, GEN1 column
Noise scaling regression (R^2 0.87 vs 0.59); scaled floors −10.60/−5.63; target floor −11.41; ISI+EQ 0.82/1.15/1.70; jitter 0.95; stacks 3.74/4.07/4.62 → +1.67/+1.34/+0.79; required 4.03 μA; TDEC proxy	`04_gen2_106g_feasibility.py`	GEN2 spec §2–6
BW window plateau −8.3/−8.4 dBm; noise-shape ≤ 0 dB; peaking/GD sweeps; 12.45 ps ripple → h_{-1} 0.477; Z_T ≥ 57 dBΩ; 162–696 μApp / 731 μA; 1.34 MHz; `verify_tia()`	`05_tia_requirements.py`	GEN2 spec §7
24-cell margin matrix; FIR isolation (0.27 dB); MRM sweep; A/B crossover 3.6 μA	`06_device_tradeoffs.py`	device trade-offs

Canvas cross-reference

- `canvases/200G-OCI-link-budget.canvas.tsx` — spec-side framing: TDEC-coupled OMA law, SRS closure-by-construction, Q table, IEEE parameter table (below), COUPE waterfall cross-check.
- `canvases/OCI-link-budget-bottom-up.canvas.tsx` — GEN1 bottom-up budget (§3 here).
- `canvases/OCI-GEN2-CPO-spec.canvas.tsx` — GEN2 budget, proposed spec, TIA requirements (§4–5 here).
- `canvases/OCI-GEN2-device-tradeoffs.canvas.tsx` — device trade study (§6 here).

Reproduction status: canvas numbers reproduced exactly or within ±0.01 dB rounding at the time the canvases were made, but the report has since adopted two deliberate accounting changes the canvases do not carry:

1. **End reflectance −24 dB for GEN1** (shared GEN1/GEN2 product line, §3.3) — the bottom-up canvas's 2.87 dB / +0.66 dB used the spec's −19 dB. Script `02` reproduces the canvas with `MPI_REFL_DB=-19`.
2. **Noise integration bandwidth** $B_n = 1.5 \times f_{3\text{dB}}$ (§2.3, register row 23) — the canvases used signal-TF shape integrals (GEN1 28.2 GHz, GEN2 64 GHz), giving GEN1 shot/RIN 0.11/0.41 (now 0.18/0.68), GEN2 RIN+shot 0.82 (now 1.16), GEN2 margins +2.00/+1.67/+1.12 (now +1.67/+1.34/+0.79), and the noise ceiling 4.37 μA / 17 pA/√Hz (now 4.03 μA / 13.6 pA/√Hz). Scripts reproduce the canvases by reverting the `BWn`/`BN_T` lines to the shape integrals (see script comments).

One accounting footnote: the spec canvas's "$i_n \le 4.4\ \mu$A → +2 dB margin" was additionally derived before the 25 fF microbump was charged to the ISI line; the current equivalents are 4.03 μA pre-bump / 3.83 μA with the bump charged. Script `04` prints both accountings.

OCI MSA vs IEEE P802.3dj (D1.3) parameter table

Parameter	OCI MSA v1.0	Cl. 181 FR4-500	Cl. 180 DR
Modulation / baud	53.125 GBd NRZ × 4λ	106.25 GBd PAM4 × 4λ	106.25 GBd PAM4 / fiber
Reach / channel IL	500 m / 2.5 dB	500 m / 3.5 dB	500 m / 3.0 dB
Penalty allocation	0.2 dB (MPI only)	3.9 dB (0.5 MPI+DGD)	3.5 dB (0.1 MPI+DGD)
Tx OMA min (worst TDEC/Q)	−3.5 dBm	+3.3 dBm	+2.2 dBm
TDEC(Q) / SEC(Q) max	3.4 / 3.4 dB	3.4 / 3.4 dB	3.4 / 3.4 dB
Rx sensitivity (SRS, worst Tx)	−6.2 dBm	−0.7 dBm	−0.9 dBm
ER min	3.5 dB	3.5 dB	3.5 dB
RIN / ORL tolerance	−138 dB/Hz @ 21.4 dB	−139 dB/Hz @ 17.1 dB	−139 dB/Hz @ 21.4 dB

Parameter	OCI MSA v1.0	Cl. 181 FR4-500	Cl. 180 DR
Tx/Rx reflectance max	−19 dB	−26 dB	−26 dB
Discrete reflectance	not specified	−25…−41 dB by count (Table 181-10)	−35 dB
BER basis	flat pre-FEC 2.4e-4	block error ratio (174A)	same

---

10. Appendix B: references and inputs

Standards and specs

- *200G OCI Optical PHY Specification v1.0* (March 2026) — `food/200G-OCI-Optical-Phy-Specification-v1.0.pdf`. Key inputs: Tx OMA min $\max(-5.5, -6.9 + \mathrm{TDEC})$ dBm, TDEC ≤ 3.4 dB, ER 3.5–4.5 dB, 500 m SMF-28 link with IL ≤ 2.5 dB and CD −0.9…+1.7 ps/nm, MPI tolerance 0.2 dB, SRS −6.2 dBm, RIN$_{\mathrm{OMA}}$ ≤ −138 dB/Hz @ 21.4 dB ORL, Rx/Tx reflectance ≤ −19 dB.
- *IEEE P802.3dj Draft 1.3* (Dec 2024), Clauses 180/181 — `food/Copy of P802d3dj draft D1p3 as shared with OIF 2024_12.pdf`. Used as the budget-structure template (penalty allocations, Table 181-10 discrete-reflectance framework, TDECQ/SRS measurement methods). The OCI MSA closely mirrors Clause 181 (800GBASE-FR4-500) recast from 106.25 GBd PAM4 to 53.125 GBd NRZ.
- Receiver/transmitter/jitter methodology (Sackinger-style): `food/lecture4_ee721_rx_analysis.pdf` (Personick sensitivity, input-referral, power penalties), `food/lecture7_ee721_tx_analysis.pdf` (ER, dispersion), `food/lecture10_ee720_jitter.pdf` (dual-Dirac total jitter).
- MPI model: Bhatt / King IEEE 802.3bs presentations — `food/bhatt_3bs_01a_0116 (1).pdf`, `food/king_01a_0116_smf (1).pdf`.

Repo data (read directly by the scripts)

- TIA characterization, 152 settings (TT corner): `LM-link-vpiphotonics/Mesa/Components/TIA/Ocelot_TIA_ADFET.vtmg_pack/Inputs/TT_TIA_DATA_ED_ADFET_20250520/` — `TT_Tia_Noise.csv` (output noise rms per setting) and `TT_Tia_TF_<key>.csv` (complex single-ended p/n-leg transfer functions and the PD column, 0–67 GHz).
- Tx driver class: `LM-link-vpiphotonics/Caribou/COUPE_models/DWDM_TxDrv/DWDM_COUPE_TxDrv.vtmg_pack/Inputs/Typ_Txdrv_NL.csv` (~1.2 V$_{\mathrm{ppd}}$ swing class; used to ground the realistic-Tx OMA case).
- Package traces (the channel CPO eliminates): `Caribou_EOE/Package/TL_TX_64G.s4p`, `TL_RX_64G.s4p` (differential S21: 1.31/0.55 dB IL at 26.6 GHz; 1.61/0.72 dB at 53.1 GHz).
- GEN2 architecture block diagram: `OCI-GEN2_Simplified.png` (copied into this folder from `mydocs/oci_msa/OCI-GEN2 Simplified.png`; reproduced in §4.1). Cascaded 2× 1-UI delay line feeding pre/main/post limiting driver slices, summed through a microbump into the MRM (laser feeds the MRM directly); mux/channel/demux, PD, second microbump, TIA; CTLE-only Rx, no DFE.
- TIA options A/B and the 60 GHz-driver trade case in §6 are user-provided device brackets (no repo file).

Nothing under `sandbox/alex` feeds the scripts; the independently produced COUPE BIDI waterfall (−11.8 dBm OMA at BER 10^{-12}) is cited once in §3 as a sanity cross-check only.